

Materials Characterization Test Report

Microstructural, Crystallographic & Mechanical Evaluation of Ti-6Al-4V

MERIDIAN ANALYTICAL LABS

Nexa Technologies Group – ISO/IEC 17025

Advanced Metallurgy & Testing Division

REPORT REF: MCR-2026-TI64-009

Sample & Client Specification

Client Entity: Apex Advanced Materials Corp.

Project Ref: Vertex Structural Engine Component

Material Grade: Titanium Ti-6Al-4V (Grade 5 / ELI)

Process Type: Directed Energy Deposition (DED)

Heat Treatment: Solution Treated & Aged (STA)
(925°C/2h/WQ + 540°C/4h/AC)

Sample ID: NEX-TI64-B09 (Coupons TC-01 to TC-05)

Batch Lot: VTX-2026-0418

Test Authorization & Standards

Date Received: July 12, 2026

Testing Dates: July 15, 2026 – July 21, 2026

Lead Engineer: Dr. Marcus Vance, Senior Metallurgist

Approved By: Dr. Elena Rostova, Lab Director

Standards Applied:

- ASTM E3-11: Metallographic Sample Prep
- ASTM E8/E8M: Tension Testing of Metallic Materials
- ASTM E112-13: Average Grain Size Determination
- ASTM E975-13: X-Ray Phase Quantification

0.2% YIELD STRENGTH

924.6 MPa

Pass (Spec ≥ 900 MPa)

TENSILE STRENGTH (σ_{UTS})

1008.2 MPa

Pass (Spec ≥ 930 MPa)

ELONGATION AT BREAK

12.8%

Pass (Spec ≥ 10.0%)

PHASE RATIO (α/β)

88.2% / 11.8%

Optimal Lamellar

Executive Summary & Compliance Declaration

Comprehensive physical, microstructural, crystallographic, and mechanical characterizations were executed on additive manufactured titanium alloy (Ti-6Al-4V) sample lot NEX-TI64-B09 submitted by Apex Advanced Materials Corp. Analytical evaluation was carried out in accordance with certified laboratory quality management protocols under ISO/IEC 17025 accreditation.

Microstructural Analysis: High-resolution FE-SEM imaging demonstrated a fully transformed $\alpha + \beta$ Widmanstätten (basket-weave) colony structure encapsulated within prior- β columnar grains oriented parallel to the additive build direction (Z-axis). The mean α -lath thickness was quantified at $1.42 \pm 0.18 \mu\text{m}$. Porosity fraction was exceptionally low (0.042%), consisting of dispersed gas entrapped micro-pores ($d < 15 \mu\text{m}$) without un-melted region defects.

Phase Quantification: X-ray diffraction (XRD) Rietveld profile refinement verified equilibrium room-temperature phase volume fractions of $88.2 \pm 0.8\%$ HCP α -titanium and $11.8 \pm 0.8\%$ BCC β -titanium, confirming effective dissolution of non-equilibrium martensitic α' phase during solution treatment and aging.

Mechanical Conformance: Uniaxial tensile testing of 5 machined coupons (TC-01 through TC-05) demonstrated mean 0.2% yield strength of 924.6 MPa, ultimate tensile strength of 1008.2 MPa, elongation at fracture of 12.8%, and area reduction of 31.4%. All mechanical metrics comfortably surpass ASTM F1472-20 and ASTM B348 Grade 5 specification thresholds.

Final Determination: Material lot NEX-TI64-B09 is formally **APPROVED** for aerospace flight hardware fabrication under Vertex Structural Program specification VTX-2026-0418.

1 Analytical Instrumentation & Experimental Methods

Metallographic specimens were sectioned using a precision diamond wafering saw, mounted in conductive bakelite, and polished down to a $0.04 \mu\text{m}$ colloidal silica finish before etching with Kroll's reagent (2% HF, 4% HNO₃, 94% H₂O) for 15 seconds.

Table 1: Summary of analytical instruments, operational parameters, and calibration standards.

Technique	Equipment Model	Operational Parameters	Calibration Standard
SEM Imaging	Zeiss Sigma 500 VP FE-SEM	Accelerating Voltage: 20.0 kV, WD: 8.5 mm, SE2 Detector	Gold On Carbon Standard (Resolution < 1.0 nm)
XRD Analysis	Rigaku SmartLab 9kW	Cu-K α ($\lambda = 1.5406 \text{ \AA}$), 40 kV/44 mA, $2\theta = 30^\circ - 80^\circ$	NIST SRM 640e Silicon Powder Standard
Tensile Testing	Instron 5985 Dual-Column 250kN	Strain Rate: 0.005 min^{-1} , Non-contact Video Extensometer	ISO 7500-1 Class 0.5 Load Cell Calibration
Density Measurement	Mettler Toledo Archimedes	Temperature-compensated Deionized Water ($\rho = 0.9982 \text{ g/cm}^3$)	Certified 100.0 g Class E2 Stainless Mass